

Mitochondrial DNA mutations accumulated in HIV-1-infected children who have an excellent virological response when exposed to long-term antiretroviral therapy

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Objectives: There is growing concern about mitochondrial DNA (mtDNA) mutations with long-term NRTI exposure in HIV-1 infected children.

Methods: Twenty-four HIV-1 infected children who started ART more than 2 years earlier who had an excellent virological response and had not changed their regimen were enrolled retrospectively. Their corresponding PBMCs in 2009 (T1), 2010 (T2) and 2013 (T3) were included. Sequencing of the entire mtDNA using next-generation sequencing revealed the spectrum of mtDNA variants.

Results: The trend showed that the number of mtDNA mutations during ART occurred as T1 < T2 < T3 ($P = 0.086$). Interestingly, the numbers of whole mtDNA mutations at T3 (median 41, range 24–62) were significantly greater than at T1 (34, 25–46, $P = 0.029$). A positive correlation was found between total mtDNA mutations and treatment time ($r = 0.352$, $P = 0.002$). During the observation period, mtDNA mutations more frequently occurred in the D-loop, cytochrome b (CYTB) and 12S rRNA regions. The heteroplasmic ratio of T3 was higher than that of T1 in CYTB and 12S rRNA ($P = 0.034$ and $P = 0.042$, respectively). High heteroplasmic population levels were found at nt 263 (A263G, D-loop) and nt 8860 (A8860G, ATPase6). A significant difference in heteroplasmy between T1, T2 and T3 occurred at nt 14783 (T14783C, CYTB, $P = 0.048$, T3 > T2 > T1).

Conclusions: Our findings reveal the spectrum of mtDNA variants in HIV-1-infected children who had an excellent virological response. mtDNA mutations accumulated during ART may play an important role in facilitating the occurrence of mitochondrial dysfunction.

Introduction

With the advent of HAART, paediatric HIV-associated morbidity and mortality have significantly decreased.^{1–3} China's free national paediatric antiretroviral therapy programme (NPATP) sponsored by the National Centre for AIDS/STD Control and Prevention (NCAIDS) was started in July 2005.⁴ China's NPATP began in six provinces in 2005 with the donation of paediatric HIV medication formulations from the Clinton Foundation and was scaled up to cover the entire country in 2006.⁵ By 2010, the paediatric ART programme had expanded to 28 provinces and autonomous regions.¹ Nonetheless, children with HIV-1 infection will inevitably be exposed to ART for an increasing period throughout post-natal

growth and development, which may lead to accumulated toxicities and emergence of severe adverse effects.⁶ NRTIs, as the backbone of ART, are responsible for significant adverse effects resulting in mitochondrial toxicity, such as lactic acidosis, hepatic steatosis, peripheral neuropathy, myopathy, cardiomyopathy, pancreatitis and lipodystrophy.^{7,8}

It is well established that the NRTI-induced inhibition of DNA polymerase γ leads to the depletion and deletion of mitochondrial DNA (mtDNA), the impairment of mtDNA and the acquisition of mtDNA point mutations, which are consequences of mitochondrial dysfunction.^{9,10} Given this effect of NRTI therapy on mtDNA

replication, it is plausible that NRTI treatment may also lead to increased mtDNA mutations. In adults, reports have suggested alterations in mtDNA over time in individuals receiving HAART. The first longitudinal study that found an increase in mtDNA mutations with NRTI therapy came from studying mutations in PBMCs.¹¹ Several studies have suggested an increase in mtDNA mutations in HIV-1-infected patients.^{12–14} In contrast, some studies have failed to replicate the above findings: they observed no accumulation of mtDNA mutations in peripheral blood samples, although it is possible that low-level mutations might have been missed.^{15,16}

The effects of NRTIs on mitochondrial structure and function in HIV-infected adults are well recognized; however, data are more limited in paediatric populations. Paediatric studies assessing the impact of exposure to NRTIs or the use of NRTIs with lower mitochondrial toxicity are urgently required. In our previous studies, we have reported mitochondrial toxicity in PBMCs of children from the Chinese national paediatric HAART cohort. That study indicated that the NRTI treatment group exhibited significant mtDNA loss, and HIV-1-infected children in this cohort on long-term HAART could suffer severe mitochondrial injury.¹⁷ In the same cohort, we also observed an increased trend of blood mtDNA mutations in HIV-1-infected children who have an excellent virological response when exposed to long-term antiretroviral therapy, compared with age-matched healthy children.¹⁸ Given the above results, there is growing concern about the mtDNA mutations associated with long-term NRTI exposure in HIV-1-infected children. Here, we carried out a retrospective longitudinal study of blood mtDNA in HIV-infected children who were long-term ART-treated and achieved virological suppression.

Patients and methods

Study population

All patients were retrospectively enrolled from rural Shangcai County, Henan province, a subset of the Paediatric AIDS Clinical Trial Group (PACTG) in China. The first-line HAART regimens previously recommended for children with HIV in China were zidovudine or stavudine plus lamivudine plus nevirapine or efavirenz. However, in 2014, some new drugs began to gradually replace old drugs; for example, efavirenz replaced nevirapine, and abacavir replaced stavudine (which is no longer used). The observation period in this study was from 2005 to 2013. Individuals who met the following criteria were retrospectively included in this study: HIV-1-infected children who (i) started a long-term (>2 years) stavudine- or zidovudine-containing combination ART regimen, i.e. HIV-1-infected children exposed to long-term antiretroviral therapy; (ii) never changed their regimen during the observation period; (iii) had an excellent virological response to ART (namely, virological suppression, defined as viral load persistently <50 copies/mL after ART, tested every 6 months) to avoid HIV infection-associated mtDNA mutations as much as possible; and (iv) had no observed clinical or laboratory toxicities. At the end of 2013, 24 paediatric patients (median age 9 years; range 6–14 years) from Shangcai County met the above criteria (see Table S1, available as [Supplementary data](#) at JAC Online). Among them, 10 patients were included from our previous study.¹⁸ Their corresponding PBMCs in 2009 (T1), 2010 (T2) and 2013 (T3) were examined to explore mtDNA mutations under the long-term effect of NRTI exposure for HIV-1-infected children in this study.

Ethics

The study was approved by the ethics review committee of Beijing YouAn Hospital, and all parents of the above subjects signed written consent forms

to participate in the research study prior to blood and clinical data collection.

DNA extraction and long-range PCR

PBMCs were isolated from fresh whole blood and stored at -80°C prior to extraction of DNA for the experiments. Total DNA was extracted from PBMCs using the QIAamp DNA Blood Mini Kit (QIAGEN, Hilden, Germany) according to the manufacturer's instructions. The entire mitochondrial genome was obtained by long-range PCR amplification of two overlapping fragments, in which each covered half of the mitochondrial genome as described previously.¹⁸ Products were purified using a QIAquick Gel Purification Kit (QIAGEN, CA, USA).

Next-generation sequencing and bioinformatics analysis

Purified amplicon pairs generated from long-range PCR were combined at equal concentrations prior to the generation of the amplicon libraries. Libraries were prepared using a multiplexed barcoding method and sequencing was conducted on an Illumina HiSeq 2000 platform. Paired-end reads of all samples were distinguished and sorted by barcode sequences during data analysis. Sorted reads were aligned to the reference mitochondrial genome using the Burrows–Wheeler Aligner (BWA). SNPs were then mapped against the revised Cambridge reference sequence of human mtDNA based on the mpileup files generated with SAMtools. Variant calls were filtered using a quality score of ≥ 20 and minimum coverage ≥ 10 .

Statistical analysis

Comparisons of the variable differences between T1, T2 and T3 were calculated using the Kruskal–Wallis test or χ^2 tests. Differences in two variables were analysed by the Mann–Whitney test. Correlation analyses were performed using a Spearman's rank test. Statistical significance was set at a two-sided P value of <0.05 . All statistical analyses were performed using the SPSS software package (version 17.0, SPSS Inc., Chicago, IL, USA).

Results

Numbers of mtDNA mutations at T3 were significantly greater than at T1

The entire mitochondrial genomes at T1, T2 and T3 during ART were analysed. The numbers of mtDNA mutations at the three time-points were compared with the reference mtDNA. The median (range) total numbers of mtDNA variation sites at the three time-points were 34 (25–46) in T1, 37 (24–59) in T2 and 41 (24–62) in T3, respectively. Comparisons between the three timepoints were made with the Kruskal–Wallis test. The trend showed that the numbers of mtDNA mutations occurred as $T1 < T2 < T3$, although no significance was observed ($P = 0.086$). However, the numbers of entire mtDNA mutations at T3 were significantly greater than at T1 (Figure 1a, $P = 0.029$).

Correlation between the numbers of mtDNA mutations and treatment time

The potential correlation between the numbers of entire mtDNA mutations and the treatment time was analysed with Spearman's rank order correlation test. We found that the numbers of entire mtDNA mutations were positively correlated with treatment time (Figure 1b, $r = 0.352$, $P = 0.002$). These results suggest that blood mtDNA mutations may accumulate in long-term ART-treated,

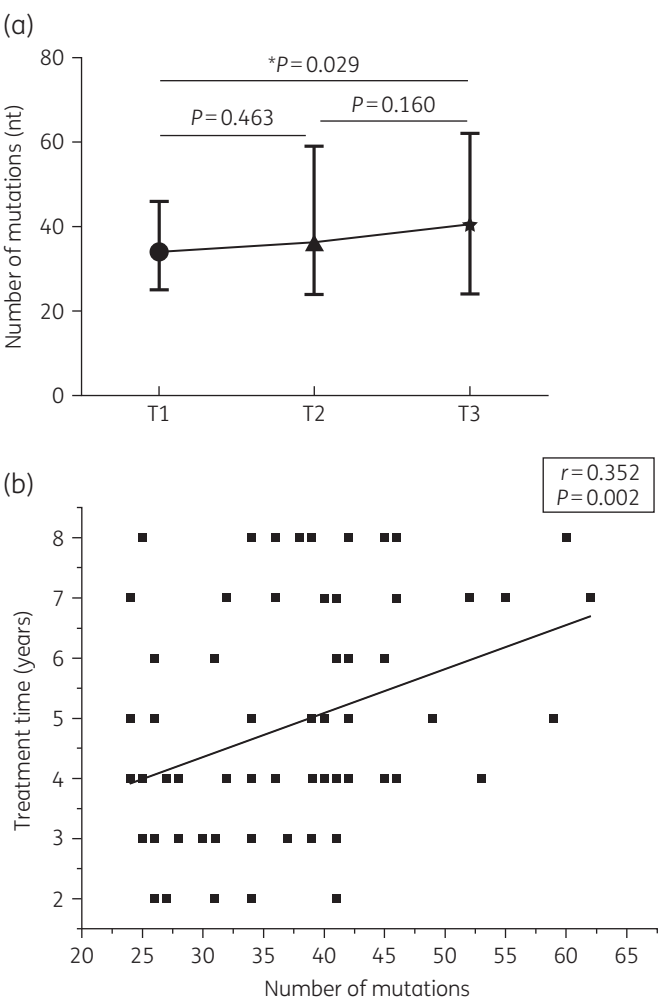


Figure 1. (a) The median number of total nt variation sites from the entire blood mitochondrial genome was compared between T1, T2 and T3. The numbers of entire mtDNA mutations at T3 were significantly greater than at T1. Error bars represent the median and range. (b) Correlation between the numbers of total mtDNA mutations and the treatment time. The numbers of entire mtDNA mutations were positively correlated with treatment time. *P* values (two-sided) and *r* values are based on Spearman's rank test.

HIV-1-infected children who had an excellent virological response to ART as the treatment time increases.

D-loop, cytochrome b and 12S rRNA were the top three mtDNA mutation regions during ART

We individually analysed the numbers of mutations in distinct regions of the mtDNA at T1, T2 and T3, including the mtDNA mutation ratio from the mtDNA D-loop, tRNA/ribosomal RNA and the 17 mtDNA-encoded polypeptides (Figure 2a). In most regions, it was shown that the numbers of mtDNA mutations at T3 were higher than those at T1, though significant differences were only found in cytochrome b (CYTB) ($P=0.044$). We then internally compared differences in the numbers of mutations of the 17 polypeptides at each treatment timepoint using the Kruskal–Wallis test according to the mean rank of the numbers of mutations for each

polypeptide. Consistently, the D-loop, CYTB and 12S rRNA had the highest frequency of mutations, and ND4L was the region with the lowest frequency at the three timepoints (Figure 2b). For mutation sites (Figure 2c) in the D-loop, CYTB and 12S rRNA, although the mutation ratio between T1, T2 and T3 at a specific site was not significantly different, the above results indicated that specific mtDNA mutation hotspot regions exist during ART.

mtDNA heteroplasmy levels changed with ART

The mtDNA heteroplasmy levels of each patient were measured at T1, T2 and T3 (Table S2). We then focused on the D-loop, 12S rRNA, CYTB and ATPase6 regions to detect nucleotide heteroplasmy in mtDNA from the three treatment times. The heteroplasmic status was measured in all four regions and it ranged from 0% to 100% (Figure 3a and Table S3). The heteroplasmic ratio at T3 was higher in CYTB and 12S rRNA than at T1 ($P=0.034$, $P=0.042$, respectively, Figure 3b). Specifically, within the mtDNA nucleotide heteroplasmic sites, during ART, a high heteroplasmic population level was found at nt 263 (in the D-loop, T1=95.83%, T2=95.65%, T3=86.96%) and at nt 8860 (in ATPase6, T1=66.67%, T2=82.61%, T3=85.71%). Specifically, the heteroplasmy level at these two nucleotides in the 24 patients at T1, T2 and T3 is shown in Figure 3(c), which showed higher or lower heteroplasmy levels in T3 both at nt 263 and nt 8860, and it varied by patient. The significant difference in heteroplasmy between T1, T2 and T3 occurred at nt 14783 (in CYTB, $P=0.048$, T3 > T2 > T1). As for the total numbers of heteroplasmic sites, no significant correlation was found between the number of heteroplasmic sites and treatment time ($r=0.154$, $P=0.196$).

Potential correlation between heteroplasmic ratio in the four mtDNA regions and the corresponding numbers of mutations

The potential correlation between heteroplasmic ratio in the four mtDNA regions and the corresponding numbers of mutations was analysed with the Spearman rank order correlation test. As shown in Figure 4, we found that the heteroplasmic ratios in CYTB and 12S rRNA were positively correlated with the corresponding numbers of mutations ($r=0.343$, $P=0.003$; $r=0.329$, $P=0.005$), but in ATPase6 it was inversely correlated with the corresponding numbers of mutations ($r=-0.327$, $P=0.006$). No significant correlation was observed between the heteroplasmic ratio in the D-loop and the corresponding numbers of mutations ($r=0.180$, $P=0.131$).

Discussion

Although HIV-associated morbidity and mortality have significantly decreased with effective ART, allowing HIV-infected children to live decades longer than before, the long-term effects of ART exposure must be considered.^{6,7} The potential adverse effects of NRTIs may affect growth and development owing to children's long-term exposure and because children may be more vulnerable than adults to potential side effects.^{19,20} Given the effect of NRTI therapy on mtDNA replication, it is plausible that NRTI treatment may also lead to increased mtDNA mutations. Currently, the potential effect of ART on the paediatric mitochondrial genome remains unclear, especially in the genetic mutation features of

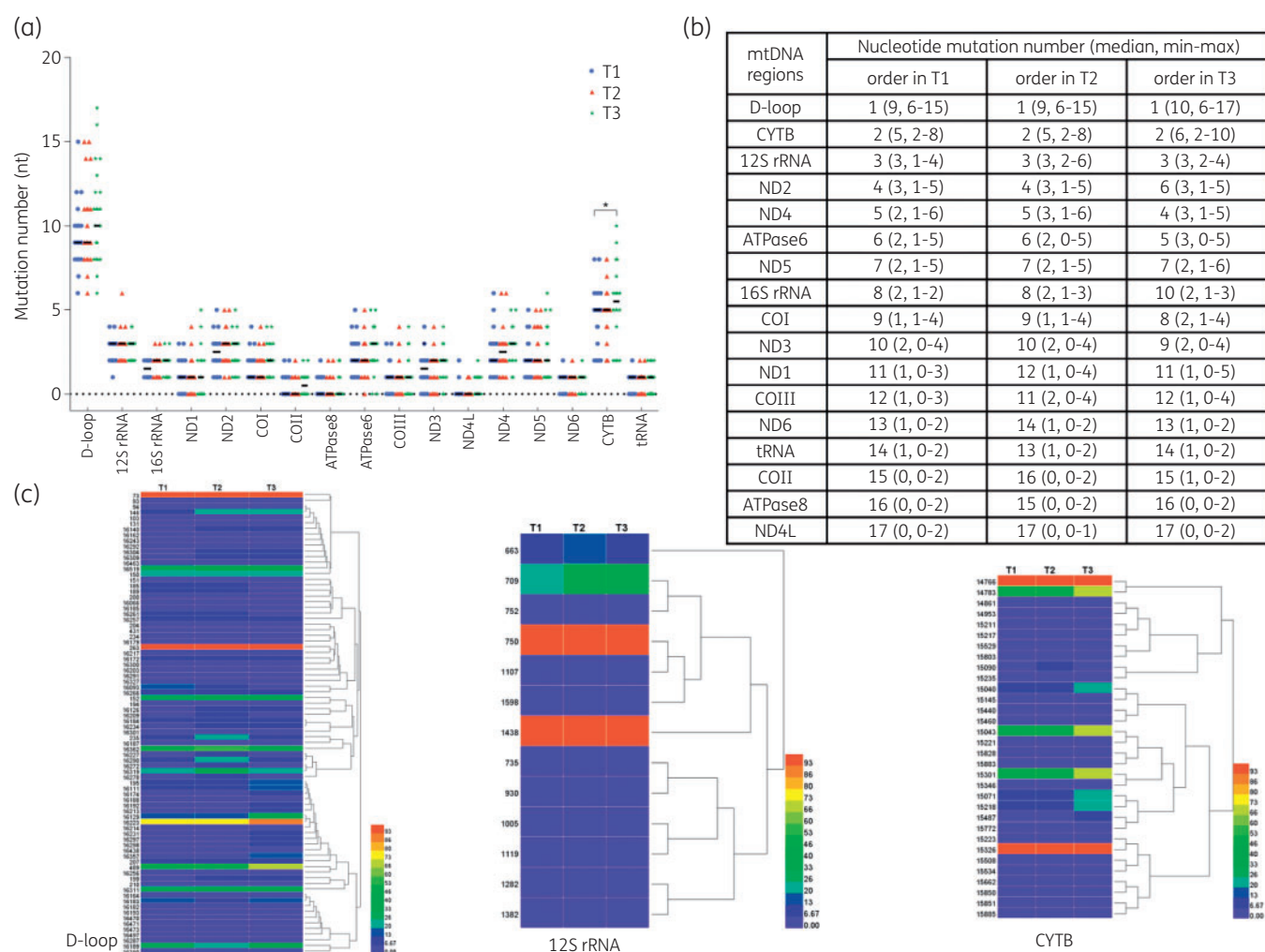


Figure 2. (a) Scatter dot plot depicting the distribution of numbers of mtDNA mutations in distinct regions at T1, T2 and T3 when compared with reference mtDNA. Comparisons between the timepoints were made with the Mann-Whitney *U*-test and $P < 0.05$ was regarded as statistically significant. Short lines for each plot represent the median. (b) Orders of the number of nucleotide mtDNA mutations in the D-loop, tRNA/ribosomal RNA and the mtDNA-encoded polypeptides. The D-loop, CYTB and 12S rRNA had the highest frequency of mutations, and ND4L was the region with the lowest frequency at all three timepoints. (c) Specific mtDNA mutation sites in the D-loop, CYTB and 12S rRNA during ART. No significantly different mutation ratios were found among T1, T2 and T3 at specific sites.

entire mtDNA derived from long-term ART-treated, HIV-1-infected children. HIV infection may be associated with mtDNA gene mutations in patients, and the hierarchy of polymerase γ *in vitro* inhibition varies by antiretroviral regimen (zalcitabine > didanosine > stavudine > zidovudine/lamivudine > abacavir/tenofovir),^{21,22} which prompted us to include HIV-1-infected children who were treated with long-term ART, who never changed their regimen and had an excellent virological response to ART, as the research subjects to avoid the effect of HIV infection as much as possible. Our previous cross-sectional studies have indicated that mitochondrial toxicity and mtDNA genetic mutations occurred in long-term ART-treated, HIV-infected children. The above results gave us some pointers, which prompted our study of mtDNA mutations in these children during the course of HAART. Here we conducted retrospective research into the mtDNA mutations of long-term ART-treated, HIV-infected children using next-generation sequencing.

Mutations and deletions in mtDNA are associated with a wide range of human diseases.⁸ NRTIs may cause an increased risk of mtDNA mutations and may provide conditions favourable for mtDNA mutations. It is plausible that one of the mechanisms may be via the acceleration of the clonal expansion of pre-existing (age-associated) mtDNA mutations.²² Martin *et al.*¹¹ reported the accumulation of multiple mutations in mtDNA extracted from the peripheral blood during ART. Wu *et al.*²³ reported that stavudine treatment is associated with a significant increase in the frequency of D-loop mutations. Li *et al.*¹⁴ observed a high frequency of mitochondrial DNA mutations in HIV-infected individuals who underwent treatment. The results of Jitratkosol *et al.*¹³ suggest that HIV and/or HAART exposure are associated with an increased prevalence of AC/TG mtDNA mutations in mothers and showed a similar tendency in infants exposed during pregnancy. In our previous study, it was shown that the total numbers of blood mtDNA mutations, in HIV-1-infected children who had an excellent virological

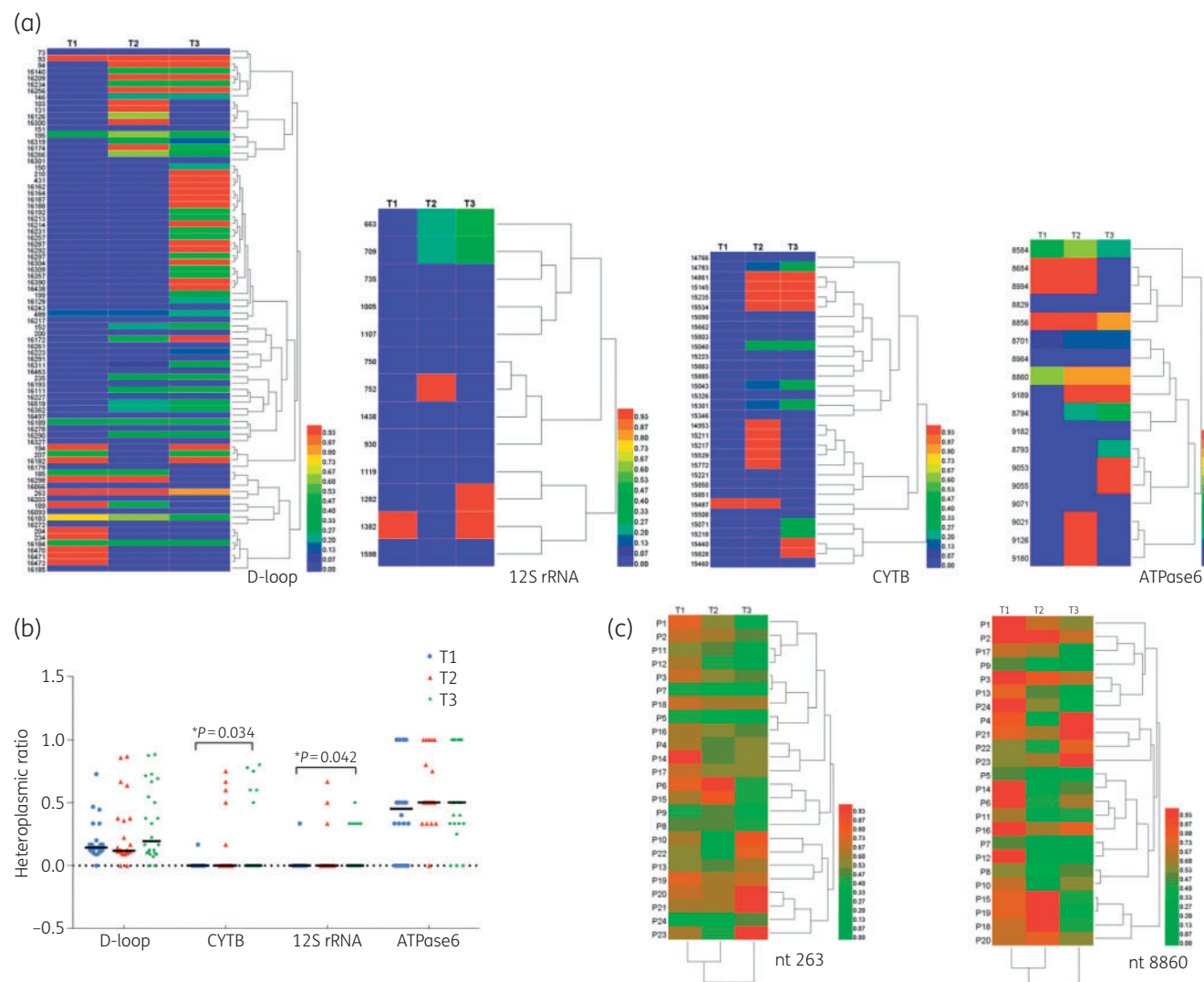


Figure 3. (a) mtDNA mutation sites with heteroplasmy in the D-loop, CYTB, 12S rRNA and ATPase6 during ART. A significant difference in heteroplasmy between T1, T2 and T3 occurred at nt 14783 (T14783C in CYTB, $P = 0.048$, $T3 > T2 > T1$). (b) Heteroplasmy ratio between T1, T2 and T3 in the D-loop, CYTB, 12S rRNA and ATPase6. The heteroplasmy ratio of T3 was higher than that of T1 in CYTB and 12S rRNA ($P = 0.034$, $P = 0.042$, respectively). (c) The heteroplasmy level of two mtDNA nt sites of 24 HIV-1-infected children at T1, T2 and T3 timepoints.

response to long-term ART, were three times greater than those in controls.¹⁸ However, Ortiz *et al.*¹⁵ found no longitudinal mtDNA sequence changes in HIV-infected individuals with or without lipodystrophy. McComsey *et al.*¹⁶ also reported several mutations in NRTI-treated patients, but the number of subjects who developed clinical mitochondrial toxicities was small, the median duration of follow-up was short, and the authors did not precisely identify the significance of the mutations identified. Therefore, prospective longitudinal studies are required in paediatric cohorts to further establish the impact of HAART and NRTIs on mitochondrial toxicities. In this study, during ART of long-term ART-treated HIV-1-infected children, we found that the median number of blood mtDNA mutations at the T3 timepoint was significantly greater than at the T1 timepoint, a trend shown as $T1 < T2 < T3$. Specifically, the

number of entire mtDNA mutations was positively correlated with treatment time. This finding supports the accumulation of mtDNA mutations as a major driver behind the development of ART. As the accumulation of mtDNA mutations has been linked with ageing and age-associated diseases, this may raise long-term concerns for HIV- and HAART-exposed children.¹³

We then assessed the mitochondrial genome for longitudinal mtDNA changes during ART in paediatric HIV-1-infected patients. Hotspot mutation regions in the derived blood mtDNA sequences between the three treatment timepoints were found at the D-loop, CYTB and 12S rRNA, which were the top three mutation regions at each treatment timepoint, consistent with our previous study.¹⁸ Although no significant mtDNA mutation sites were found at the three timepoints, some of the mutation sites accumulated

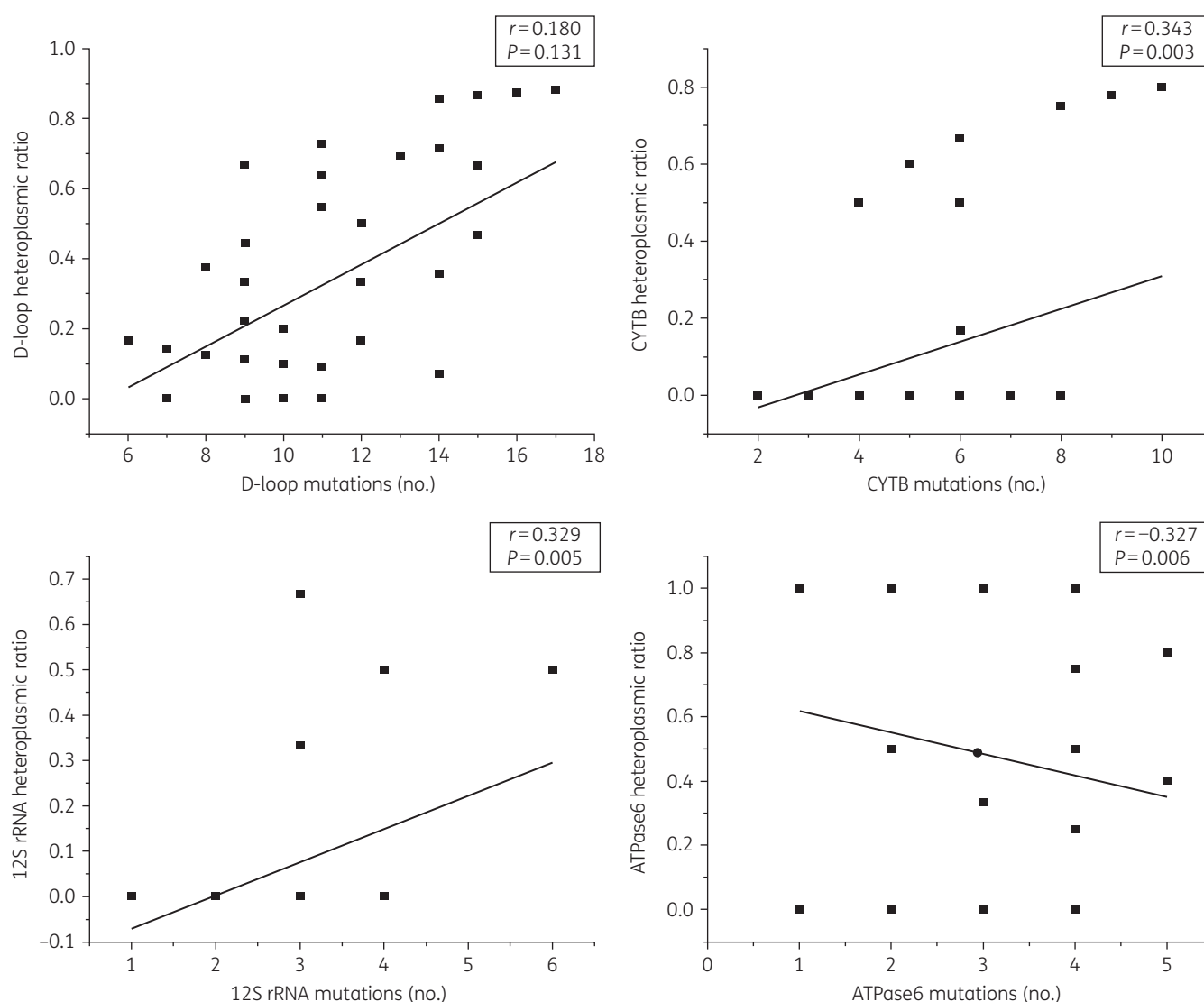


Figure 4. Correlation between the heteroplasmic ratio in the D-loop, CYTB, 12S rRNA and ATPase6 regions and the corresponding numbers of mutations. *P* values (two-sided) and *r* values are based on Spearman's rank test.

with treatment time. The presence of these mutation sites during ART in paediatric HIV-1-infected patients may contribute to the development of future mitochondrial diseases.

In particular, mtDNA point mutations may occur somatically and accumulate with ageing, coexisting with the WT, a condition known as heteroplasmy.²⁴ Under specific circumstances, the clonal expansion of mutant mtDNA may occur within single cells, causing a wide range of severe human diseases when the mutant overcomes the WT. Improved sequencing techniques and the widespread investigation of mtDNA in human and animal tissues have shown that heteroplasmy is much more common than initially believed, particularly considering the ageing of post-mitotic tissues where mutations accumulate with time.^{25,26} In adults, reports have suggested alterations in mtDNA heteroplasmy over time in individuals receiving HAART.^{11,16} We have shown a higher heteroplasmic ratio at T3 in CYTB and 12S rRNA than at T1. However, in our study, a high level of heteroplasmy only occurs at

some nt sites. For example, A263G mutations (in the D-loop) and A8860G mutations (in ATPase6) always displayed higher heteroplasmy levels at each treatment timepoint, and patients showed an upward or downward trend with time. Only the heteroplasmy level in mtDNA at nt 14783 (T14783C) increased with treatment time. The A263G variant has been associated with muscle pathology, auditory neuropathy, hypertrophic cardiomyopathy, idiopathic cardiopathy and infantile cardiomyopathy. The A8860G variant was reported in subfertility, abdominal aortic aneurysm, Leber hereditary optic neuropathy (LHON), dystonia, colorectal cancer and hypertrophic cardiomyopathy.²⁷⁻²⁹ T14783C is probably a polymorphism of the Northeast Thais and may not be associated with cholangiocarcinoma.³⁰

In our longitudinal study, mtDNA mutations accumulated in children exposed to ART over time, which may play an important role in facilitating the occurrence of mitochondrial dysfunction. Higher mtDNA heteroplasmy levels were maintained

through to the end of the study period at some nucleotide sites, whereas the only difference was found at nt 14783 (T14783C) over time. Ongoing investigations will clarify the impact of these regimens on mitochondrial toxicity. Skeletal muscles, the best indicator of mitochondrial toxicity, were difficult to assess in children. Therefore, assessments of ART-associated mitochondrial toxicity have frequently used PBMCs as the cellular source, especially for children.³¹ In the observed period, mitochondrial damage-related mutations existed, perhaps at the molecular level, in the absence of a clinical phenotype. In conclusion, this study revealed a spectrum of mtDNA variants in HIV-1-infected children during ART. Our results identified a synonymous change, the T14783C variant, whose heteroplasmy level increased with treatment time. This study indicated that for paediatric patients who are receiving long-term ART, in addition to regular monitoring of viral load and CD4+ cell counts, mtDNA mutations should also be used as a monitoring indicator. Although we demonstrated statistical differences in terms of the presence of mtDNA mutations among HIV-1-infected children during ART, there are still some limitations to our study. It remains difficult to maintain a well-matched control group (healthy children) over the same follow-up period in order to determine the normal ratios of mutations. Owing to the long observation period and the problem of drug replacement at a certain timepoint, the study's population size limits the robustness of the conclusions. Furthermore, the mtDNA mutations under current first-line paediatric ART regimens should be addressed in the near future.

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Transparency declarations

None to declare.

Author contributions

D. C. and D. L. conceived of the study. Y. O., F. W., L. Q., K. L., Y. D., X. G., S. W., L. P., M. L. and F. Z. performed the experiments and/or participated in the experimental design. Y. O. and D. C. performed the statistical analysis. D. C., Y. O., D. L., F. W. and L. Q. wrote and edited the manuscript. All authors read and approved the final manuscript.

Supplementary data

Tables S1 to S3 appear as [Supplementary data](#) at JAC Online.

References

- 1 Zhao Y, Li C, Sun X et al. Mortality and treatment outcomes of China's National Pediatric antiretroviral therapy program. *Clin Infect Dis* 2013; **56**: 735–44.
- 2 Sánchez JM, Ramos AJ, Fernández DMS et al. Impact of highly active antiretroviral therapy on the morbidity and mortality in Spanish human immunodeficiency virus-infected children. *Pediatr Infect Dis J* 2003; **22**: 863–7.
- 3 Chearskul P, Chokephaibulkit K, Chearskul S et al. Effect of antiretroviral therapy in human immunodeficiency virus-infected children. *J Med Assoc Thai* 2005; **88**: S221–31.
- 4 Zhao Y, Sun X, He Y et al. Progress of the National Pediatric Free Antiretroviral Therapy program in China. *AIDS Care* 2010; **22**: 1182–8.
- 5 Ma Y, Zhang F, Zhao Y et al. Cohort profile: the Chinese national free antiretroviral treatment cohort. *Int J Epidemiol* 2010; **39**: 973–9.
- 6 Foster C, Lyall H. HIV and mitochondrial toxicity in children. *J Antimicrob Chemother* 2008; **61**: 8–12.
- 7 Lewis W, Dalakas MC. Mitochondrial toxicity of antiviral drugs. *Nat Med* 1995; **1**: 417–22.
- 8 Park CB, Larsson NG. Mitochondrial DNA mutations in disease and aging. *J Cell Biol* 2011; **193**: 809–18.
- 9 Lewis W, Day BJ, Copeland WC. Mitochondrial toxicity of NRTI antiviral drugs: an integrated cellular perspective. *Nat Rev Drug Discov* 2003; **2**: 812–22.
- 10 Côté HC, Brumme ZL, Craib KJ et al. Changes in mitochondrial DNA as a marker of nucleoside toxicity in HIV-infected patients. *N Engl J Med* 2002; **346**: 811–20.
- 11 Martin AM, Hammond E, Nolan D et al. Accumulation of mitochondrial DNA mutations in human immunodeficiency virus-infected patients treated with nucleoside-analogue reverse-transcriptase inhibitors. *Am J Hum Genet* 2003; **72**: 549–60.
- 12 Payne BA, Wilson IJ, Hateley CA et al. Mitochondrial aging is accelerated by anti-retroviral therapy through the clonal expansion of mtDNA mutations. *Nat Genet* 2011; **43**: 806–10.
- 13 Jitratkosol MH, Sattha B, Maan EJ et al. Blood mitochondrial DNA mutations in HIV-infected women and their infants exposed to HAART during pregnancy. *AIDS* 2012; **26**: 675–83.
- 14 Li M, Foli Y, Liu Z et al. High frequency of mitochondrial DNA mutations in HIV-infected treatment-experienced individuals. *HIV Med* 2017; **18**: 45–55.
- 15 Ortiz M, Poloni ES, Furrer H et al. No longitudinal mitochondrial DNA sequence changes in HIV-infected individuals with and without lipodystrophy. *J Infect Dis* 2011; **203**: 620–4.
- 16 McComsey G, Bai RK, Maa JF et al. Extensive investigations of mitochondrial DNA genome in treated HIV-infected subjects: beyond mitochondrial DNA depletion. *J Acquir Immune Defic Syndr* 2005; **39**: 181–8.
- 17 Liu K, Sun Y, Liu D et al. Mitochondrial toxicity studied with the PBMC of children from the Chinese national pediatric highly active antiretroviral therapy cohort. *PLoS One* 2013; **8**: e57223.
- 18 Ouyang Y, Qiao L, Liu K et al. Mitochondrial DNA mutations in blood samples from HIV-1-infected children undergoing long-term antiretroviral therapy. *Mutat Res Genet Toxicol Environ Mutagen* 2016; **805**: 1–6.
- 19 McComsey GA, Leonard E. Metabolic complications of HIV therapy in children. *AIDS* 2004; **18**: 1753–68.
- 20 Jao J, Abrams EJ. Metabolic complications of in utero maternal HIV and antiretroviral exposure in HIV-exposed infants. *Pediatr Infect Dis J* 2014; **33**: 734–40.

- 21** Lim SE, Copeland WC. Differential incorporation and removal of antiviral deoxynucleotides by human DNA polymerase γ . *J Biol Chem* 2001; **276**: 23616–23.
- 22** Payne BA, Gardner K, Chinnery PF. Mitochondrial DNA mutations in ageing and disease: implications for HIV? *Antivir Ther* 2015; **20**: 109–20.
- 23** Wu Y, Li N, Zhang T et al. Mitochondrial DNA base excision repair and mitochondrial DNA mutation in human hepatic HuH-7 cells exposed to stavudine. *Mutat Res* 2009; **664**: 28–38.
- 24** Carelli V, Maresca A, Caporali L et al. Mitochondria: biogenesis and mitophagy balance in segregation and clonal expansion of mitochondrial DNA mutations. *Int J Biochem Cell Biol* 2015; **63**: 21–4.
- 25** Larsson NG. Somatic mitochondrial DNA mutations in mammalian aging. *Annu Rev Biochem* 2010; **79**: 683–706.
- 26** Khrapko K, Turnbull D. Mitochondrial DNA mutations in aging. *Prog Mol Biol Transl Sci* 2014; **127**: 29–62.
- 27** Houshmand M, Montazeri M, Kuchekian N et al. Is 8860 variation a rare polymorphism or associated as a secondary effect in HCM disease? *Arch Med Sci* 2011; **7**: 242–6.
- 28** Güney AI, Javadova D, Kirac D et al. Detection of Y chromosome microdeletions and mitochondrial DNA mutations in male infertility patients. *Genet Mol Res* 2012; **11**: 1039–48.
- 29** Kumar R, Venkatesh S, Kumar M et al. Oxidative stress and sperm mitochondrial DNA mutation in idiopathic oligoasthenozoospermic men. *Indian J Biochem Biophys* 2009; **46**: 172–7.
- 30** Muisuk K, Silsirivanit A, Imtawil K et al. Novel mutations in cholangiocarcinoma with low frequencies revealed by whole mitochondrial genome sequencing. *Asian Pac J Cancer Prev* 2015; **16**: 1737–42.
- 31** Saitoh A, Fenton T, Alvero C et al. Impact of nucleoside reverse transcriptase inhibitors on mitochondria in human immunodeficiency virus type 1-infected children receiving highly active antiretroviral therapy. *Antimicrob Agents Chemother* 2007; **51**: 4236–42.